

BLM2041 Signals and Systems

Syllabus

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BLM2041 Signals and Systems

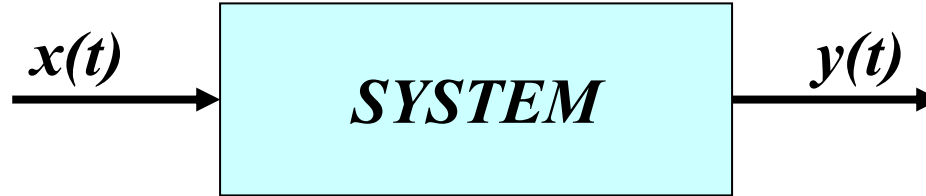
Sampling & Aliasing

LECTURE OBJECTIVES

- SAMPLING can cause ALIASING
 - Sampling Theorem
 - Sampling Rate $> 2(\text{Highest Frequency})$
- Spectrum for digital signals, $x[n]$
 - Normalized Frequency

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s} + 2\pi\ell$$

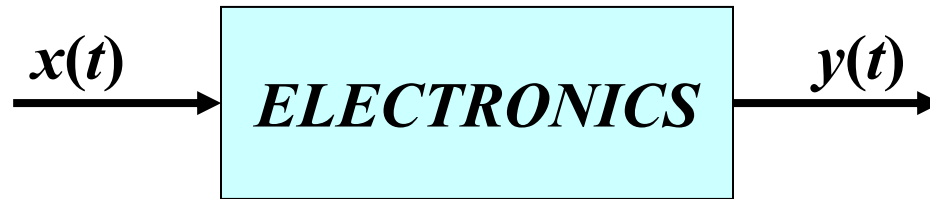

SYSTEMS Process Signals



- PROCESSING GOALS:
 - Change $x(t)$ into $y(t)$
 - For example, more BASS
 - Improve $x(t)$,
 - e.g., image deblurring
 - Extract information from $x(t)$

System IMPLEMENTATION

- ANALOG/ELECTRONIC:
 - Circuits: resistors, capacitors, op-amps

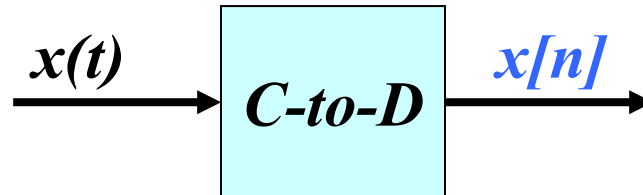


- DIGITAL/MICROPROCESSOR
 - Convert $x(t)$ to numbers stored in memory



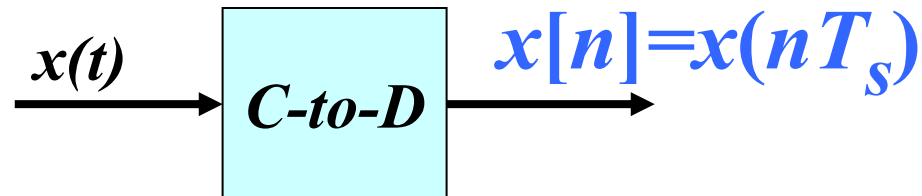
SAMPLING $x(t)$

- SAMPLING PROCESS
 - Convert $x(t)$ to numbers $x[n]$
 - “ n ” is an integer;
 - $x[n]$ is a sequence of values
 - Think of “ n ” as the storage address in memory
- UNIFORM SAMPLING at $t = nT_s$
 - IDEAL: $x[n] = x(nT_s)$



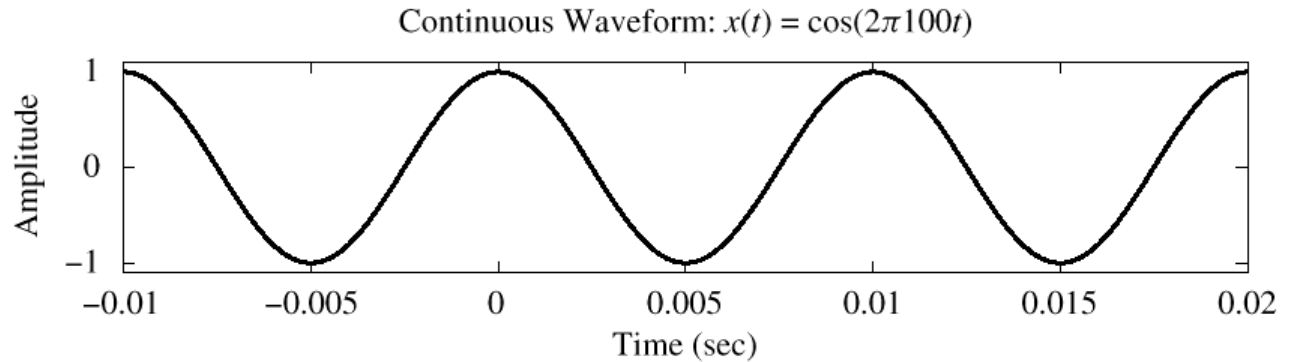
SAMPLING RATE, f_s

- SAMPLING RATE (f_s)
 - $f_s = 1/T_s$
 - NUMBER of SAMPLES PER SECOND
 - $T_s = 125$ microsec $\rightarrow f_s = 8000$ samples/sec
 - UNITS ARE HERTZ: 8000 Hz
- UNIFORM SAMPLING at $t = nT_s = n/f_s$
 - IDEAL: $x[n] = x(nT_s) = x(n/f_s)$

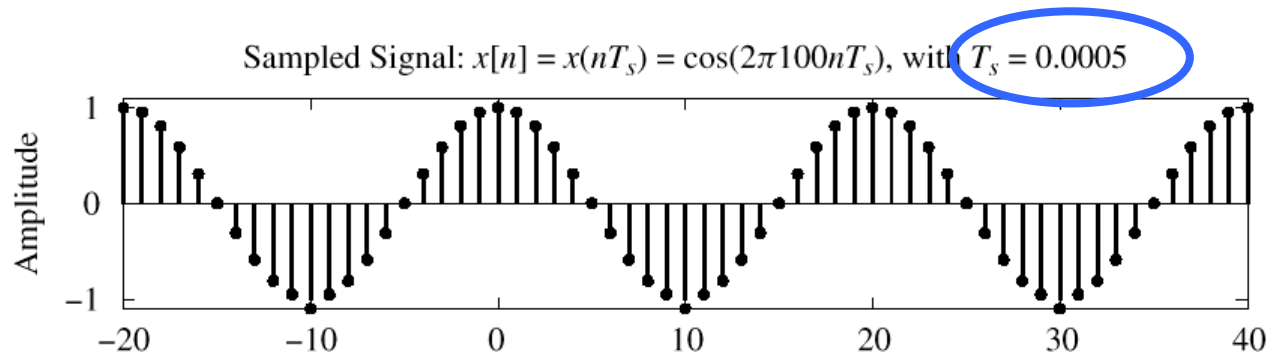


SAMPLING RATE, f_s

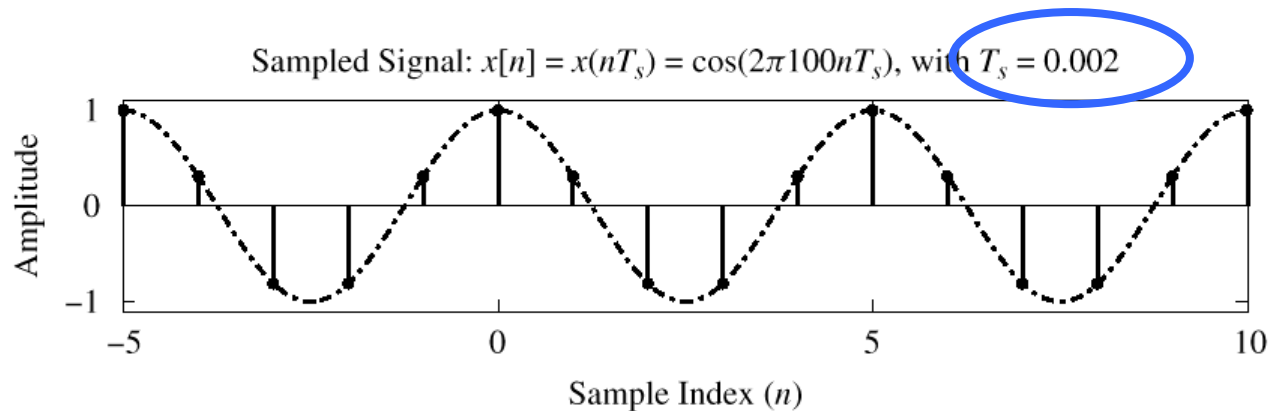
$$f = 100\text{Hz}$$



$$f_s = 2\text{ kHz}$$



$$f_s = 500\text{Hz}$$



SAMPLING THEOREM

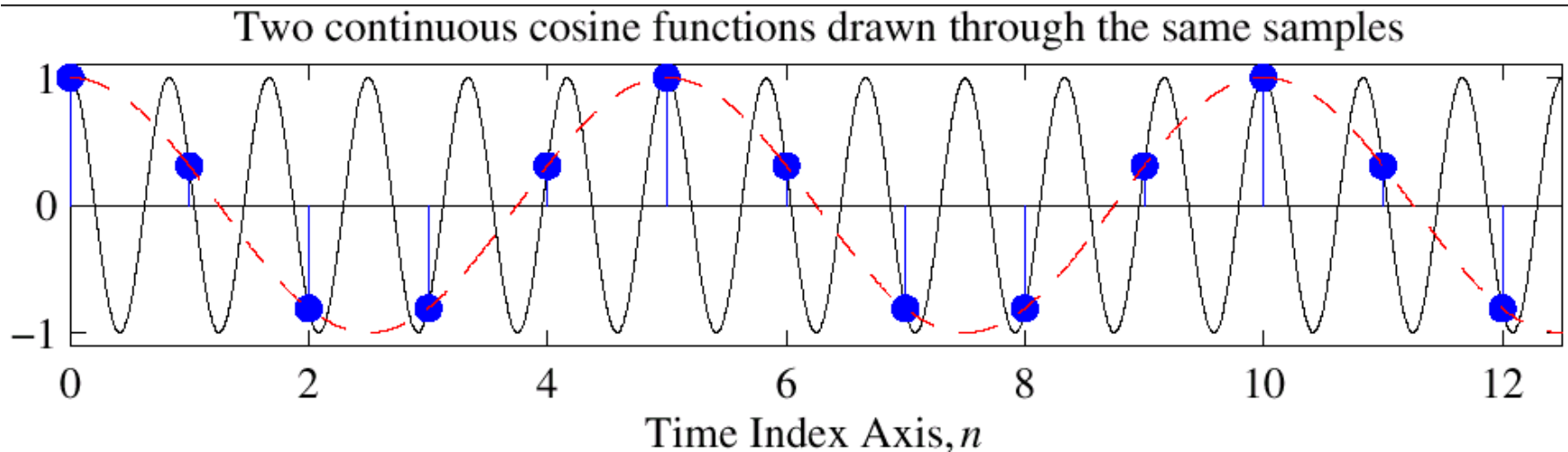
- HOW OFTEN ?
 - DEPENDS on FREQUENCY of SINUSOID
 - ANSWERED by SHANNON/NYQUIST Theorem
 - ALSO DEPENDS on “RECONSTRUCTION”

Shannon Sampling Theorem

A continuous-time signal $x(t)$ with frequencies no higher than $f_{\max}$ can be reconstructed exactly from its samples $x[n] = x(nT_s)$, if the samples are taken at a rate $f_s = 1/T_s$ that is greater than $2f_{\max}$.

Reconstruction? Which One?

Given the samples, draw a sinusoid through the values



$$x[n] = \cos(0.4\pi n)$$

When n is an integer
 $\cos(0.4\pi n) = \cos(2.4\pi n)$

STORING DIGITAL SOUND

- $x[n]$ is a SAMPLED SINUSOID
 - A list of numbers stored in memory
- EXAMPLE: audio CD
- CD rate is 44,100 samples per second
 - 16-bit samples
 - Stereo uses 2 channels
- Number of bytes for 1 minute is
 - $2 \times (16/8) \times 60 \times 44100 = 10.584$ Mbytes

DISCRETE-TIME SINUSOID

- Change $x(t)$ into $x[n]$

- **DERIVATION**

$$x(t) = A \cos(\omega t + \varphi)$$

$$x[n] = x(nT_s) = A \cos(\omega nT_s + \varphi)$$

$$x[n] = A \cos((\omega T_s)n + \varphi)$$

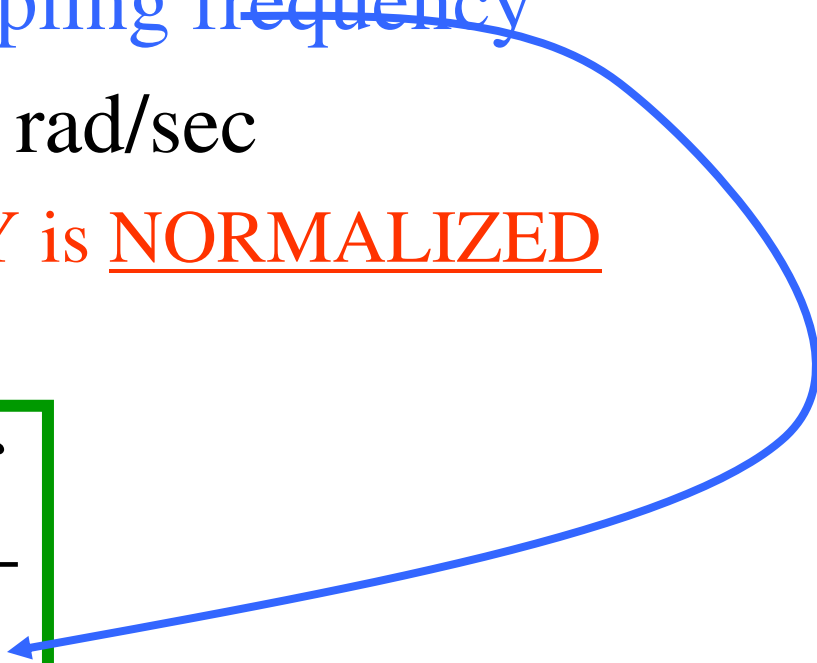
$$x[n] = A \cos(\hat{\omega}n + \varphi)$$

$$\hat{\omega} = \omega T_s = \frac{\omega}{f_s}$$

DEFINE DIGITAL FREQUENCY

DIGITAL FREQUENCY

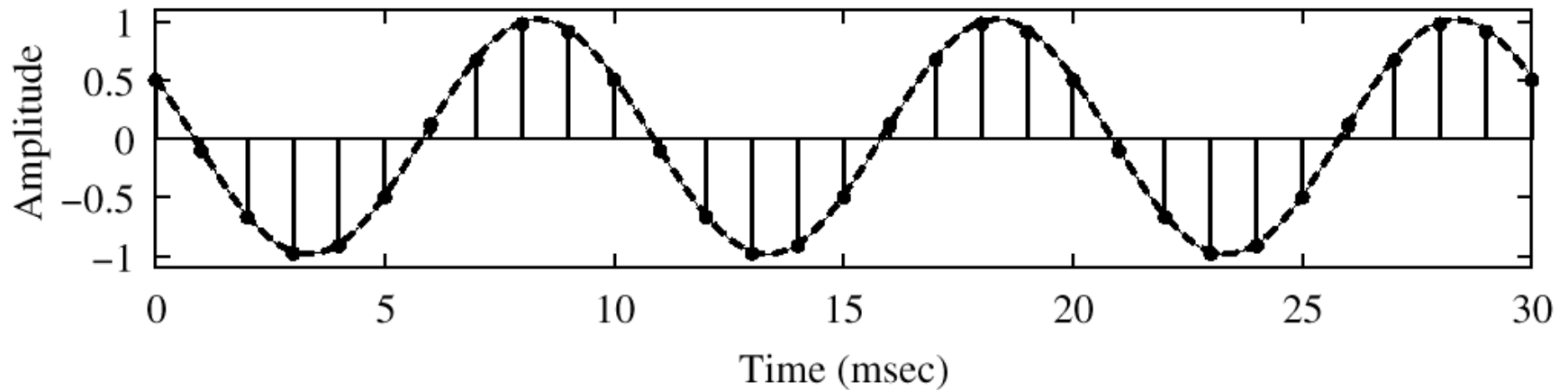
- Digital frequency $\hat{\omega}$ VARIES from **0** to **2π** , as f varies from **0** to the **sampling frequency**
- UNITS are radians, not rad/sec
 - **DIGITAL FREQUENCY is NORMALIZED**

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$$


SPECTRUM (DIGITAL)

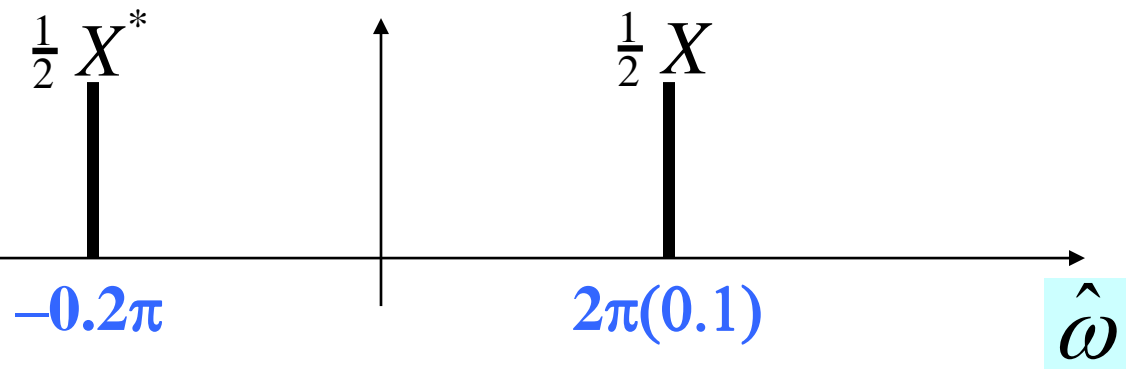
$$x[n] = A \cos(2\pi(100)(n/1000) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 1$ msec (1000 Hz)



$$\hat{\omega} = 2\pi \frac{f}{f_s}$$

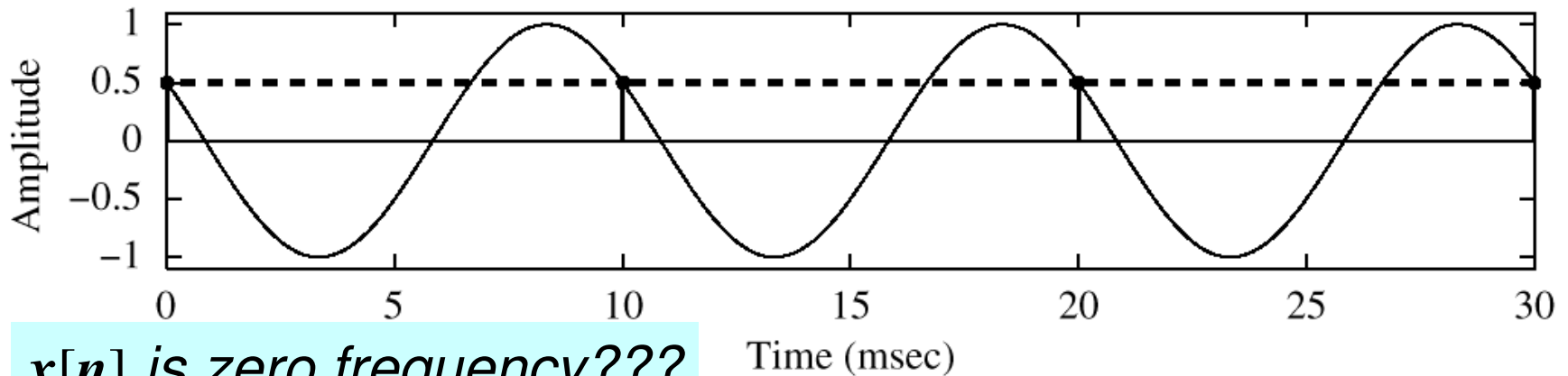
$$f_s = 1 \text{ kHz}$$



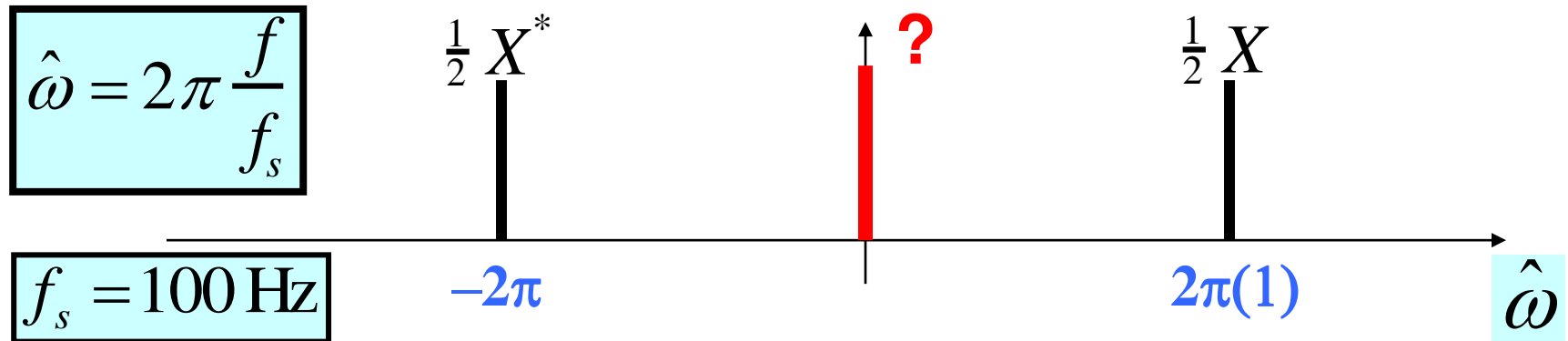
SPECTRUM (DIGITAL) ???

$$x[n] = A \cos(2\pi(100)(n/100) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 10$ msec (100 Hz)



$x[n]$ is zero frequency???



The REST of the STORY

- Spectrum of $x[n]$ has more than one line for each complex exponential
 - Called ALIASING
 - MANY SPECTRAL LINES
- SPECTRUM is PERIODIC with period = 2π
 - Because

$$A \cos(\hat{\omega}n + \varphi) = A \cos((\hat{\omega} + 2\pi)n + \varphi)$$

ALIASING DERIVATION

- Other Frequencies give the same $\hat{\omega}$

$$x_1(t) = \cos(400\pi t) \quad \text{sampled at } f_s = 1000\text{Hz}$$

$$x_1[n] = \cos(400\pi \frac{n}{1000}) = \cos(0.4\pi n)$$

$$x_2(t) = \cos(2400\pi t) \quad \text{sampled at } f_s = 1000\text{Hz}$$

$$x_2[n] = \cos(2400\pi \frac{n}{1000}) = \cos(2.4\pi n)$$

$$x_2[n] = \cos(2.4\pi n) = \cos(0.4\pi n + 2\pi n) = \cos(0.4\pi n)$$

$$\Rightarrow x_2[n] = x_1[n]$$

$$2400\pi - 400\pi = 2\pi(1000)$$

ALIASING DERIVATION-2

- Other Frequencies give the same $\hat{\omega}$

If $x(t) = A \cos(2\pi(f + \ell f_s)t + \varphi)$

$$t \leftarrow \frac{n}{f_s}$$

and we want : $x[n] = A \cos(\hat{\omega}n + \varphi)$

$$\text{then : } \hat{\omega} = \frac{2\pi(f + \ell f_s)}{f_s} = \frac{2\pi f}{f_s} + \frac{2\pi \ell f_s}{f_s}$$

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s} + 2\pi \ell$$

ALIASING CONCLUSIONS

- ADDING f_s or $2f_s$ or $-f_s$ to the FREQ of $x(t)$ gives exactly the same $x[n]$
 - The samples, $x[n] = x(n/f_s)$ are EXACTLY THE SAME VALUES
- GIVEN $x[n]$, WE CANNOT DISTINGUISH f_o FROM $(f_o + f_s)$ or $(f_o + 2f_s)$

NORMALIZED FREQUENCY

- DIGITAL FREQUENCY

Normalized Radian Frequency

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s} + 2\pi\ell$$

Normalized Cyclic Frequency

$$\hat{f} = \hat{\omega}/(2\pi) = f T_s = f/f_s$$

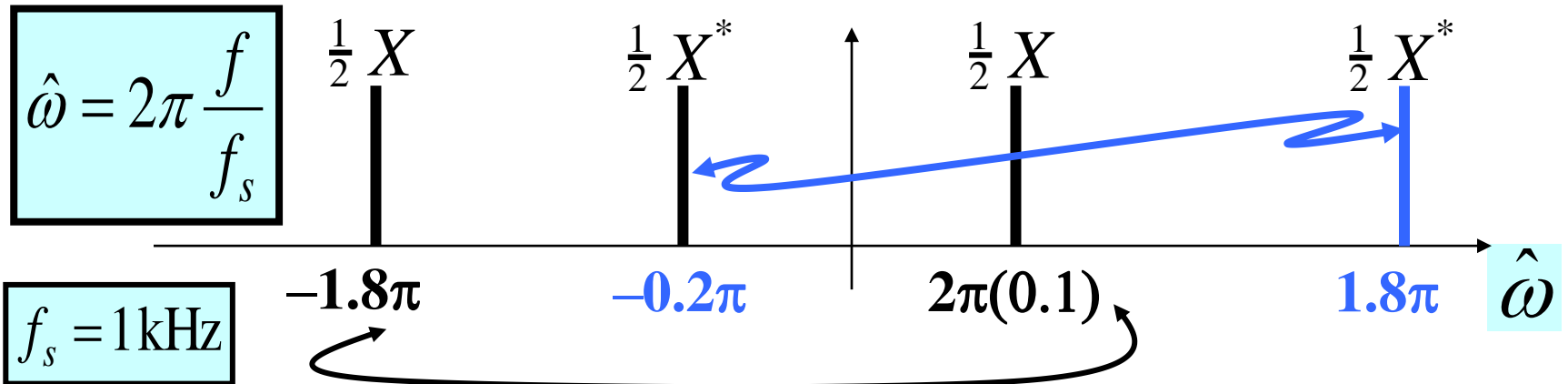
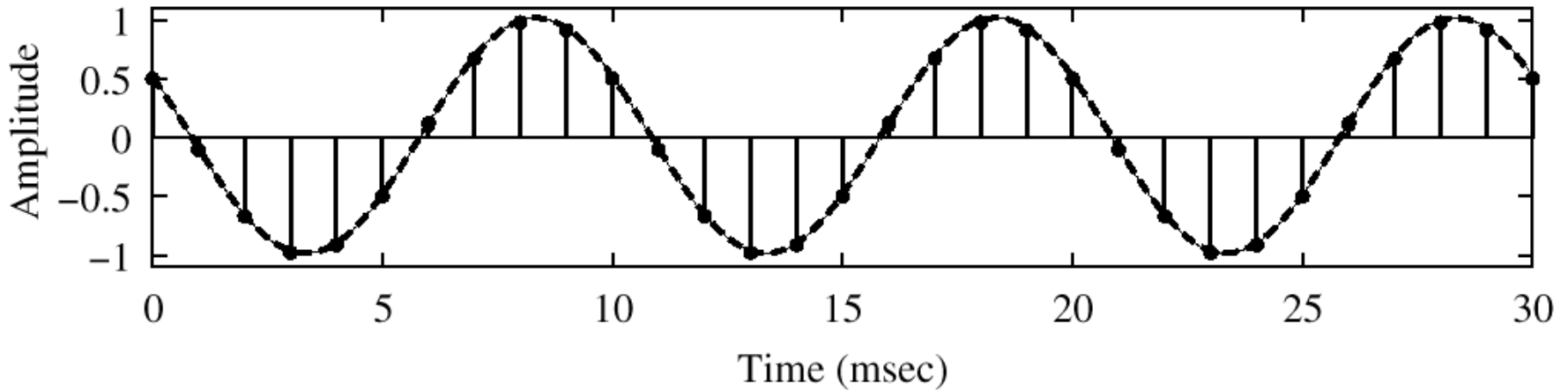
SPECTRUM for $x[n]$

- PLOT versus NORMALIZED FREQUENCY
- INCLUDE ALL SPECTRUM LINES
 - ALIASES
 - ADD MULTIPLES of 2π
 - SUBTRACT MULTIPLES of 2π
 - FOLDED ALIASES
 - ALIASES of NEGATIVE FREQS

SPECTRUM (MORE LINES)

$$x[n] = A \cos(2\pi(100)(n/1000) + \varphi)$$

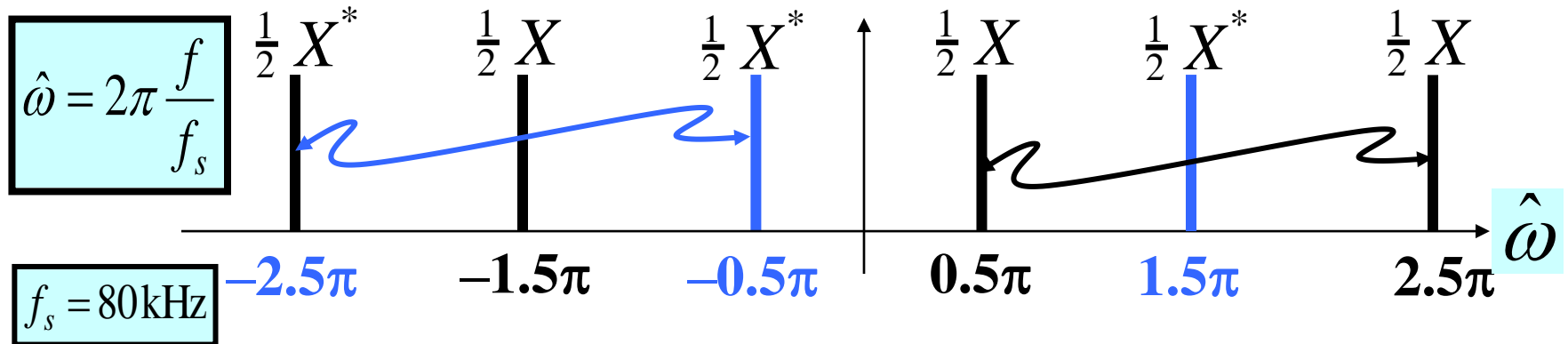
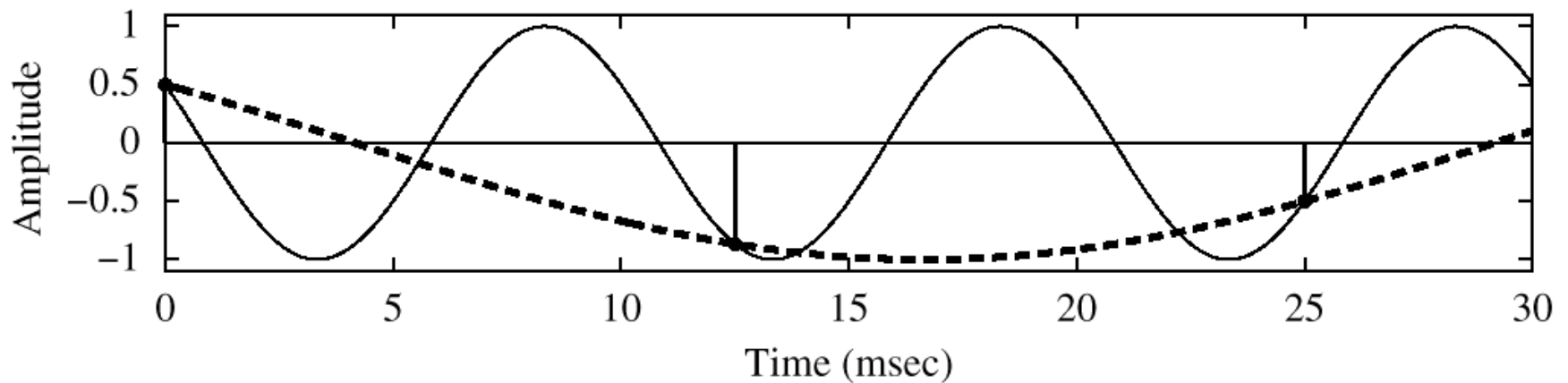
100-Hz Cosine Wave: Sampled with $T_s = 1$ msec (1000 Hz)



SPECTRUM (ALIASING CASE)

$$x[n] = A \cos(2\pi(100)(n/80) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 12.5$ msec (80 Hz)



SPECTRUM (FOLDING CASE)

$$x[n] = A \cos(2\pi(100)(n/125) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 8$ msec (125 Hz)

